

Stellenbosch University

Department of Industrial Engineering

Non-linear Optimisation 414

Tutorial 5: 8 May 2026 (on §4.3.2–§5)

1. Maximise the function

$$f(x_1, x_2) = -x_1^3 + 3x_1 - x_2^2$$

by performing three iterations of particle swarm optimisation (Algorithm 4.7), starting out from a randomly generated population of five candidate solutions having initial positions

$$\begin{bmatrix} 0.3 \\ -0.2 \end{bmatrix}, \begin{bmatrix} -0.9 \\ 0.5 \end{bmatrix}, \begin{bmatrix} -0.4 \\ 0.5 \end{bmatrix}, \begin{bmatrix} 0.9 \\ 0.7 \end{bmatrix}, \text{ and } \begin{bmatrix} -0.4 \\ -0.5 \end{bmatrix},$$

each with zero velocity initially. Adopt the algorithmic hyperparameters $\omega = 0.8$, $C^1 = C^2 = 0.2$, $\rho^1 = 0.5$ and $\rho^2 = 0.5$ and apply velocity clamping with maximum velocity vector $\mathbf{V}_{max} = [0.25 \ 0.25]$ and minimum velocity vector $\mathbf{V}_{min} = [-0.25 \ -0.25]$. Work accurately to three decimal places and produce a table of algorithmic output similar in structure to Table 4.7.

2. Find three real numbers whose sum is 12 and the sum of whose squares is as small as possible. Use the method of Lagrange multipliers to obtain your answer.
3. A package in the shape of a rectangular box can be mailed at standard price by a postal service if the sum of its length and girth (the perimeter of a cross-section perpendicular to its length) is at most 270 cm. Use the Karush-Kuhn-Tucker necessary conditions for local optimality to find the dimensions of a package of largest volume that can be mailed at standard price. Show all your calculations.
4. Use the method of Lagrange multipliers to prove that the rectangle with maximum area that has a given parameter p is, in fact, a square. You can ignore the domain constraints for this question.